Does Multi-Task Learning Work for Crystals? A Systematic Study and a Knowledge-Guided Mixture-of-Experts

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Abstract

Crystal property prediction is essential for accelerating materials discovery across applications in energy, catalysis, and electronics. While recent approaches have made significant progress on this problem, they universally follow a *single-task learning* (STL) paradigm, training one model per property. The cost of characterizing a material thus scales linearly with the number of properties, and the physical coupling among properties is ignored. Multi-task learning (MTL) is the principled alternative, yet it has not been systematically explored for crystalline materials. In this paper, we present, to the best of our knowledge, the first systematic study of MTL for crystal multi-property prediction, covering representative scalarization, gradient-manipulation, and Mixture-of-Experts (MoE) methods on three major benchmarks (SNUMAT, JARVIS-DFT, and Materials Project) spanning 18 properties. Our study reveals a pronounced efficiency-performance trade-off among existing MTL methods. Scalarization methods do not perform reliably across datasets, while gradient-manipulation methods incur substantial computational costs without consistently achieving better accuracy. MoE, although architecturally the best fit for scaling to many properties, underdelivers because its purely data-driven routing is blind to task relations. Motivated by these observations, we further analyze inter-property relationships from both physical-chemical principles and label-space correlations, and encode the resulting physical knowledge into MoE routing through a hierarchical grouping strategy. The resulting model, *GroupMoE*, is a plug-and-play architecture that routes each property through a group-specific sub-MoE complemented by shared experts. It significantly outperforms existing MTL methods across all three benchmarks, achieving the lowest overall performance degradation relative to STL, measured by $\Delta m\%$ (e.g., 1.92 on SNUMAT and 25.60 on JARVIS-DFT), while training over 3× faster than separate per-property models.

Keywords

Multi-property prediction, Mixture-of-experts, Hierarchical grouping strategy

1 Introduction

Material property prediction, estimating electronic, thermodynamic, structural, and magnetic characteristics from a material's atomic structure, is a cornerstone of AI-driven materials discovery [2, 36, 42]. Though density functional theory (DFT) has provided significant advances, its computational cost limits scalability [23]. Thus, recent machine learning models, such as Graph Neural Networks (GNNs) and transformers, have been explored for predicting material properties [18, 29, 33, 39, 56, 57]. Despite their effectiveness, these models universally follow a *single-task learning (STL) paradigm*, training one model per property. Meanwhile, materials databases track multiple properties per material, and real-world materials design requires simultaneous estimates of many of them for a full characterization. Under STL, per-property training scales linearly in compute and storage and becomes prohibitive as property catalogs expand. Moreover, material properties are physically coupled through the shared atomic structure, yet independent per-property training provides no mechanism to exploit this coupling. There is therefore a compelling need for *multi-property prediction* models that simultaneously estimate multiple properties while exploiting their underlying physical relationships.

A principled framework for multi-property prediction is multi-task learning (MTL). In the broader machine learning literature, MTL methods largely fall into three families. *Scalarization-based* methods, such as linear scalarization (LS) and uncertainty weighting (UW) [24], optimize a weighted combination of task losses, but suffer from gradient conflict when tasks have competing objectives [55]. *Gradient-manipulation* methods such as MGDA and CAGrad [10, 31] mitigate this issue by explicitly resolving per-task gradient conflicts, but require computing all task gradients at every iteration, incurring overhead that scales poorly as the number of properties grows. A third, *architectural* direction is the Mixture-of-Experts (MoE) [12, 48], which improves representation flexibility through input-dependent routing over a sparse subset of experts, mitigating severe gradient conflict without computing per-task gradients. However, to the best of our knowledge, MTL has *not* been systematically explored in the context of crystal property prediction: it remains unclear how these paradigms behave on materials data, and which of them, if any, is suited to the domain.

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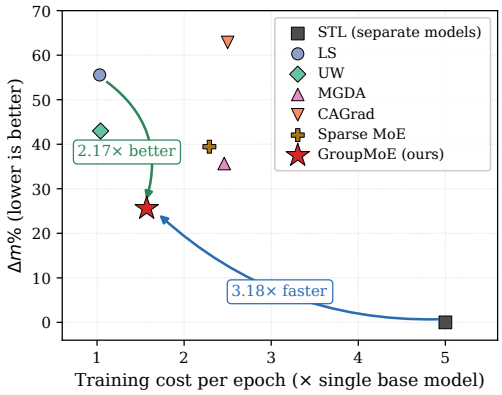


Figure 1: Efficiency-performance trade-off of multi-property prediction on JARVIS-DFT: overall performance drop relative to STL ($\Delta m\%$) versus training cost per epoch relative to a single base model. Points closer to the lower-left corner indicate a more favorable trade-off.

Answering these questions is not a straightforward transfer of general-domain MTL methods, because crystal multi-property prediction presents distinct domain challenges. Target properties span heterogeneous physical categories, including electronic, thermodynamic, structural, and magnetic quantities, and exhibit highly non-uniform relationships: some are strongly coupled, whereas others are nearly independent. The benchmarks also vary substantially in scale, label availability, and task type, including both regression and classification objectives. Moreover, it remains unclear whether MTL methods developed primarily for vision and language can be effectively transferred to crystal property prediction while preserving its domain-specific physical structure. These unique characteristics motivate three key research questions:

- **RQ1 (*Landscape*):** *How do representative MTL paradigms perform when applied to crystal property prediction?*
- **RQ2 (*Paradigm Choice*)**: *Which paradigm provides the most favorable balance of accuracy, efficiency, and scalability?*
- **RQ3 (*Knowledge Injection*)**: *How can scientific domain knowledge be incorporated to improve multi-property prediction?*

To answer **RQ1** and **RQ2**, we systematically evaluate representative scalarization, gradient-manipulation, and MoE methods, together with STL reference, on major benchmarks in the field (SNUMAT [25], JARVIS-DFT [8], and Materials Project [15]), which cover 18 properties (Section 3). **Three key findings emerge.** *First*, the efficiency-performance trade-off in this domain is substantial, and additional computation does not necessarily recover the performance gap. As Figure 1 shows on the accuracy-cost plane of JARVIS-DFT, every generic method remains far from the desirable lower-left corner: scalarization trains at roughly the cost of a single model but retains a large overall gap relative to STL, while gradient manipulation requires several times more computation yet may still underperform the least expensive baseline. *Second*, MTL performance depends strongly on the composition of the property set. On SNUMAT, which is dominated by strongly correlated bandgap variants, simple uncertainty weighting already approaches STL performance. In contrast, JARVIS-DFT and Materials Project combine bandgaps with more weakly coupled energetic, structural, and magnetic properties. As a result, the overall gap becomes substantially larger, and the best-performing method varies across datasets. *Third*, though MoE offers the most favorable balance of efficiency and scalability by addressing task interference through sparse routing, standard sparse MoE remains suboptimal and can even underperform simple scalarization methods. A key reason is that its routing is learned purely from data and does not encode scientific knowledge about which properties are physically related. This motivates the next step of our study: *Can MoE routing be enhanced with scientific domain knowledge to improve multi-property prediction while preserving interpretability?*

To address **RQ3**, we exploit a distinctive advantage of the materials domain: unlike vision or NLP, where task relations are ambiguous, crystalline property prediction has *a priori* task relations grounded in physics. For example, bandgaps computed under different DFT functionals probe the same electronic structure and are strongly related. We therefore analyze inter-property relationships from both physical-chemical principles and label-space Pearson correlations (Section 4.1). It shows that properties governed by related physical mechanisms exhibit strong statistical correlations, while properties governed by distinct physics remain weakly coupled. As a first step toward knowledge-informed MoE routing, we encode the domain knowledge through a hierarchical grouping algorithm that combines physical-chemical initialization with label-space correlation refinement. Building on this grouping, we propose **GroupMoE**, a modular architecture that partitions experts into dedicated sub-MoEs, routes each property to its corresponding group-specific sub-MoE, and complements them with shared experts. It is also *plug-and-play*, integrating seamlessly into existing GNN- and transformer-based backbones such as PotNet and ComFormer [29, 57]. Across the three benchmarks, GroupMoE attains the lowest $\Delta m\%$ among all evaluated MTL methods (1.92, 25.60, and 35.99 on SNUMAT, JARVIS-DFT, and Materials Project, respectively). On SNUMAT it approaches single-task accuracy at about 4× lower training cost, and on JARVIS-DFT it is 3.18× cheaper than per-property training while more than halving the $\Delta m\%$ of linear scalarization. In conclusion, since GroupMoE differs only in the injected task structure, its gains directly quantify the value of scientific knowledge in MoE routing. Overall, our study provides a first empirical map of MTL for crystal multi-property prediction and points to knowledge-informed routing as a general recipe for efficient, science-aware MoE design in materials and beyond.

2 Background and Study Design

We first define the object of our study: the prediction tasks and datasets (Section 2.1-2.2), the MTL method families under comparison (Section 2.3), and the evaluation protocol (Section 2.4). Detailed related works are discussed in Appendix A.

2.1 Problem Setup and Tasks

Material property prediction has been widely studied using both physics-based and deep-learning methods. Early approaches, such

Table 1: Statistics of the representative benchmark datasets in the crystalline property prediction field.

Dataset	Structure files	Structure-Description pairs				Properties
		Total	Train	Validation	Test	
SNUMAT [25]	10,481	10,481	8,385	1,048	1,048	7
JARVIS-DFT [8]	55,722	55,722	44,578	5,572	5,572	5
Materials Project [15, 20]	139,367	139,227	111,381	13,923	13,923	8
Total	205,570	205,430	164,344	20,543	20,543	20

as Coulomb matrices [11, 44], effectively model ionic and metallic materials but struggle with generalization due to permutation invariance. Deep-learning methods provide more flexible representations [16, 22, 52], with recent work increasingly modeling crystals as 3D atomic graphs. Representative approaches include SchNet [45] and CGCNN [56] for geometric and periodic interactions, ALIGNN [7] for angular information, Matformer [58] for periodic attention, and PotNet [29] for physics-informed interatomic features. More recent architectures, including iComFormer, eComFormer, Crystalformer, CrystalFramer, and ReciNet [18, 39, 50, 57], further improve invariant, equivariant, periodic, and long-range representations. Given a crystal structure, multi-property prediction aims to simultaneously estimate a set of target properties $\mathcal{T} = \{1, \ldots, T\}$ with a single model.

2.2 Datasets

We evaluate on three widely used materials science benchmarks: SNUMAT [25], JARVIS-DFT [8], and Materials Project [15]. Together, these datasets cover 18 material properties, including 15 regression tasks and 3 classification tasks, spanning electronic, energetic, structural, and magnetic properties with labels derived from different density functional theory (DFT) settings and physical definitions. The three datasets differ substantially in scale and task composition (Table 1). Materials Project provides the most comprehensive coverage with up to 139,367 samples across 8 properties; JARVIS-DFT contributes 55,722 samples spanning 5 fundamental properties with specialized functionals; and SNUMAT provides 10,481 samples with complete annotations across 7 properties, including both bandgap values and gap-type classifications. This multi-dataset, multi-task setup presents substantial challenges due to varying data distributions and sizes, missing task labels across datasets, and the need to simultaneously optimize for both regression and classification objectives with distinct physical interpretations. Detailed per-property statistics, physical meanings (Table 5), and label distributions are provided in Appendix B.

2.3 MTL Method Families

We compare representative methods from the three MTL families and also benchmark single-task learning (STL) as a reference.

STL reference. Single-task learning trains one independent model for each property, without sharing parameters or information across tasks. It therefore acts as a per-task oracle baseline but incurs the highest total training and storage cost, requiring T separate models for T properties.

Scalarization. Scalarization methods convert multiple task objectives into a single training objective through a weighted combination of task losses. The standard approach, linear scalarization (LS), optimizes a weighted or averaged sum of task losses [5]. Although efficient, LS often suffers from imbalanced training as tasks converge at different rates. To alleviate this issue, Kendall et al. [24] weights individual loss functions based on homoscedastic uncertainties, while Liu et al. [32] adjusts weights based on the relative rate of loss descent to better synchronize task optimization. These methods are simple to implement and introduce essentially no computational overhead beyond training a single shared model.

Gradient manipulation. Another line of work formulates MTL as multi-objective optimization and employs the Multi-Gradient Descent Algorithm (MGDA) to mitigate gradient conflict [10, 46]. For example, CAGrad [31] constrains the update direction, while PCGrad [59] resolves conflicts by projecting gradients onto the normal plane. Nash-MTL [38] considers the gradient combination step as a bargaining game. SDMGrad [54] precisely controls the gradient conflict in each epoch with a direction-oriented mechanism, and FairGrad [3] incorporates α-fairness. However, they require computing all T task gradients at every training step, leading to substantial computational and memory overhead as the number of properties increases.

MoE-based. Mixture-of-Experts (MoE) has recently emerged as a flexible architectural approach to multi-task learning, where different inputs or tasks can activate different subsets of model parameters, reducing task interference. Sparse MoE [48] follows this architectural paradigm. More recently, MoE has been adopted for MTL model design: M^{3}ViT [12] sparsely activates task-specific experts during training to avoid conflicts between tasks, and ModSquad [6] lets experts be both cooperative and specialized through a mutual-information loss. A MoE layer consists of a set of expert networks $E_i, \forall i \in [1, N]$, and a routing network that selects a subset of experts for each input. Experts are typically implemented as multi-layer perceptrons [13]. We adopt the widely used top-K routing strategy, where the MoE output for an input x is a weighted sum over the K selected experts:

$$q = \sum_{i=1}^{N} \mathcal{R}_i(x) E_i(x),$$
$$\text{s.t. } \mathcal{R}(x) = \text{TopK}(\text{Softmax}(xW + \mathcal{N}(0,1)\text{Softplus}(xW_{\text{noise}}))), \quad (1)$$

where W and W_{noise} are router parameters, $\mathcal{N}(0,1)$ denotes standard Gaussian noise, and the TopK$(\cdot)$ operator retains only the K largest routing weights and sets the rest to zero. Routing activates only a small number of experts per input, which mitigates gradient conflict and avoids computing per-task gradients.

2.4 Evaluation Protocol

Backbones. Crystal property prediction backbones broadly follow two architectures: graph neural networks (GNNs) and graph transformers. We include representative models from both families. For the main experiments, we select ReciNet [39], a recent competitive architecture that models long-range interactions through reciprocal-space representations, as the shared backbone for all MTL methods to ensure a controlled comparison. To study the effect of base-model capacity and verify backbone-agnostic conclusions, we additionally consider SchNet, CGCNN, Matformer, PotNet, and eComFormer [29, 45, 56–58] in Section 5.1.

Table 2: Performance comparison on the SNUMAT dataset. Regression tasks are evaluated using the test MAE, while classification tasks are evaluated using balanced accuracy. $\Delta m\%$ evaluates the overall performance. GroupMoE is introduced in Section 4.

Methods	Regression tasks				Classification tasks			$\Delta m\% \downarrow$
	HSE Optical↓ eV	GGA Optical↓ eV	Bandgap GGA↓ eV	Bandgap HSE↓ eV	Is Direct HSE↑	SOC↑	Is Direct↑	
STL	0.553	0.511	0.312	0.351	0.669	0.823	0.694	
LS	0.585	0.533	0.332	0.384	0.666	0.804	**0.686**	4.26
UW	0.578	0.529	0.324	0.379	0.654	0.830	0.671	3.51
MGDA	0.735	0.777	0.474	0.644	0.582	0.748	0.565	37.30
CAGrad	0.660	0.587	0.380	0.604	0.652	0.776	0.677	19.83
Sparse MoE	0.587	0.536	0.340	0.400	**0.668**	0.831	0.684	4.94
GroupMoE (**ours**)	**0.575**	**0.518**	**0.313**	**0.373**	0.667	**0.831**	0.679	**1.92**

Table 3: Performance comparison on the JARVIS-DFT dataset. Regression tasks are evaluated using the test MAE.

Methods	FEPA↓ eV/atom	Bandgap (MBJ)↓ eV	Bandgap (OPT)↓ eV	Etotal↓ eV/atom	Ehull↓ eV	$\Delta m\%\downarrow$
STL	0.027	0.239	0.126	0.027	0.048	
LS	0.049	0.240	0.136	0.053	0.092	55.56
UW	0.044	0.219	0.113	0.052	0.084	42.98
MGDA	0.039	0.215	0.118	0.049	0.081	35.66
CAGrad	0.046	**0.212**	0.115	0.077	0.086	62.87
Sparse MoE	0.037	0.227	0.119	0.054	0.082	39.45
GroupMoE (**ours**)	**0.034**	0.213	**0.112**	**0.047**	**0.072**	25.60

Metrics. For regression tasks, we use the test Mean Absolute Error (MAE). For classification tasks, the label distribution is imbalanced (Appendix B.2); we therefore use the balanced accuracy score (BAcc) [4], which averages recall across classes. To quantify overall multi-task performance, we report **Δm**%, the average relative performance drop of a multi-task model against its single-task counterpart [35]: $\Delta m\% = \frac{1}{K}\sum_{k=1}^{K}(-1)^{\delta_k}(M_{m,k}-M_{b,k})/M_{b,k}\times 100$, where $M_{m,k}$ and $M_{b,k}$ denote the performance of the k-th task under the multi-task model m and the single-task model b, respectively, and $\delta_k = 1$ if higher values are better for task k and 0 otherwise. Lower $\Delta m\%$ is better, and $\Delta m\% > 0$ indicates an overall gap to STL. Hyperparameter settings are summarized in Appendix C.

3 A Systematic Study of MTL Methods on Multi-Property Prediction

We now answer **RQ1** and **RQ2**: how do representative MTL methods behave on crystal multi-property prediction, and which paradigm is best suited to the domain?

3.1 Overall Results

Tables 2 and 3 report the results on SNUMAT and JARVIS-DFT, and Table 9 in the appendix reports Materials Project. For completeness, the tables include our GroupMoE (introduced in Section 4), while this section focuses on the generic MTL methods. On *SNUMAT*, scalarization-based methods perform strongly, with UW and LS achieving $\Delta m\%$ values of 3.51 and 4.26, respectively. In contrast, CAGrad and MGDA yield substantially larger gaps of 19.83 and 37.30, suggesting that explicit gradient manipulation does not necessarily improve multi-property prediction in this setting. *JARVIS-DFT* is considerably more challenging, as all generic methods exhibit larger overall gaps. MGDA performs best with a $\Delta m\%$ of 35.66, followed by sparse MoE at 39.45 and UW at 42.98, whereas LS and CAGrad reach 55.56 and 62.87, respectively. Nevertheless, several methods outperform STL on strongly coupled properties. For example, CAGrad reduces the MBJ bandgap MAE from 0.239 to 0.212, demonstrating the potential for positive transfer. On *Materials Project*, UW achieves the best generic MTL result with a $\Delta m\%$ of 36.87. LS, CAGrad, and MGDA obtain comparable values of 42.08, 43.75, and 44.50, while sparse MoE performs worst at 48.51.

3.2 Observations

Observation 1 (Efficiency-performance trade-off). *Existing MTL methods exhibit an unfavorable trade-off between predictive performance and training efficiency.* As shown in Figure 1, scalarization methods operate near the cost of a single backbone but retain large performance gaps, with LS yielding $\Delta m\% = 55.56$ on JARVIS-DFT. Gradient-manipulation methods increase per-epoch training cost to 2.46–4.49× across all datasets due to per-task gradient computation, without reliably achieving better performance. In contrast, separate STL models attain $\Delta m\% = 0$ by definition but incur the highest cost (5× for JARVIS-DFT). These results show that existing approaches do not achieve favorable scaling between computational cost and predictive performance.

Observation 2 (No generic method is consistently good). *The best generic baseline changes from dataset to dataset, and the expensive methods are the least reliable.* UW is the strongest generic method on SNUMAT (3.51) and Materials Project (36.87) but falls behind on JARVIS-DFT (42.98). MGDA is the strongest generic method on JARVIS-DFT (35.66) yet the worst on SNUMAT (37.30); CAGrad spans 19.83 (SNUMAT) to 62.87 (JARVIS-DFT). This volatility is problematic: a practitioner cannot know in advance which generic method will work on a new materials dataset, and the methods with the highest compute budgets are precisely the most volatile.

Observation 3 (MoE is prevailing, but vanilla MoE underperforms). *Architecturally, MoE offers scalability for growing properties; empirically, task-agnostic routing fails.* Unlike gradient-manipulation methods, MoE mitigates task interference through input-dependent routing over a sparse subset of experts, avoiding explicit per-task

gradient computation while providing conditional model capacity and modular scalability. Yet vanilla sparse MoE is suboptimal across datasets, 4.94 on SNUMAT (behind UW and LS), 39.45 on JARVIS-DFT, and 48.51 on Materials Project. The potential reason is that its routing is purely data-driven: experts are not aligned with any meaningful tasks. Material properties span heterogeneous physical categories with non-uniform inter-task relationships, so unconstrained routing may fail to extract physically meaningful features, leading to limited interpretability and suboptimal performance.

Takeaway 1. *Generic MTL methods remain either inaccurate or expensive: LS shows high error rates, while gradient manipulation introduces substantial computational overhead without consistent performance gains.*

Takeaway 2. *Sparse MoE offers a scalable alternative, but task-agnostic routing remains suboptimal . This motivates incorporating scientifically grounded inter-property relationships into expert routing.*

4 Injecting Domain Knowledge into MoE

Building on these takeaways, we take a step further and investigate how to push the efficiency-performance boundary by improving MoE routing, answering **RQ3**. Since task-agnostic routing does not account for the property relationships, we first study the physical and statistical inter-property relationships present in the data (Section 4.1). We then formalize this process through a hierarchical grouping algorithm (Section 4.2) and incorporate the resulting task groups into MoE routing (Section 4.3).

4.1 Property Relationship Analysis

In multi-task learning, identifying task relationships is crucial for leveraging positive transfer and minimizing negative transfer. Unlike vision or NLP, where task relations are often ambiguous or purely empirical, material properties are governed by the atomic arrangement, and understanding their relationships before training is both possible and essential. We therefore analyze task relationships from two complementary perspectives: *physical-chemical principles* and *statistical label-space correlation*. For the latter, we compute pairwise Pearson correlation coefficients between training labels within each dataset

$$\rho_{ij} = \frac{\sum_{s=1}^{S}(y_i^s - \bar{y}_i)(y_j^s - \bar{y}_j)}{\sqrt{\sum_{s=1}^{S}(y_i^s - \bar{y}_i)^2}\sqrt{\sum_{s=1}^{S}(y_j^s - \bar{y}_j)^2}}, \tag{2}$$

where y_i^s / y_j^s denotes the label of property i/j for sample s and $\bar{y}_i = \frac{1}{S}\sum_{s=1}^{S} y_i^s$. The resulting matrices are visualized in Figure 4.

SNUMAT. SNUMAT contains electronic bandgap properties computed under different DFT approximations and physical definitions, leading to hierarchical task relationships. Electronic bandgaps computed with GGA and HSE describe the same physical quantity but differ in their treatment of electron-electron interactions. The distinction between electronic and optical gaps mainly arises from momentum conservation [40]. Bandgap type (direct vs. indirect) is determined by band-structure topology and is generally consistent across functionals [47]. In contrast, spin-orbit coupling (SOC) originates from relativistic interactions that scale primarily with atomic number [27], and is therefore governed by different physics than the bandgap. The label-space correlations strongly reflect this underlying physical structure. Electronic and optical bandgaps computed with the same functional show extremely high correlation ($\rho \geq 0.95$), indicating that functional choice mainly affects scale, not ranking. Across functionals, the four regression tasks form a tightly connected cluster ($\rho \geq 0.81$), while direct/indirect classification shows strong agreement across GGA and HSE ($\rho = 0.79$), reflecting consistent band topology. SOC exhibits near-zero correlation with others ($|\rho| < 0.16$), confirming its statistical independence.

JARVIS-DFT. The five properties in JARVIS-DFT reflect diverse physical origins spanning electronic structure and thermodynamic stability. Formation energy and bandgaps are often anti-correlated: stable materials (with large negative formation energy) tend to have narrow bandgaps [9]. MBJ and OPT bandgaps, despite using different DFT functionals, aim to approximate the same Kohn-Sham energy gap [51]. Total energy and Ehull are inherently linked through the thermodynamic convex hull: materials with low total energy are more likely to be close to the hull and thus more stable [49]. The label-space correlations support these physical insights. MBJ and OPT bandgaps are strongly correlated ($\rho = 0.95$). Total energy and Ehull are negatively correlated ($\rho = -0.77$): as expected, low-energy materials tend to be more stable and close to the convex hull. Formation energy and Ehull, however, show weak correlation ($\rho = 0.07$), since Ehull reflects phase-level competition beyond intrinsic stability. Similarly, bandgaps have negligible correlation with Ehull ($\rho \approx -0.1$), reinforcing that electronic structure and thermodynamic competitiveness are largely decoupled.

Materials Project. This dataset spans electronic, thermodynamic, structural, and magnetic domains, forming a rich multi-physics prediction landscape. Similarly, bandgap and formation energy per atom (FEPA) exhibit an inverse relationship. Fermi energy (Efermi), reflecting the electrochemical potential of electrons, correlates with atomic packing. Dense materials exhibit higher Fermi levels due to Pauli exclusion, raising the highest occupied states [1]. Thermodynamic quantities, including FEPA, Efinal, and Ehull, form a conceptually related triad [20]. Finally, total magnetization arises from unpaired electron spins and is highly specific to certain material classes. The correlation structure reinforces physically grounded groupings. Efermi and density show the strongest correlation ($\rho = 0.67$), consistent with atomic packing driving the electron chemical potential. Bandgap is moderately anti-correlated with formation energy ($\rho = -0.45$), Fermi energy ($\rho = -0.43$), and density ($\rho = -0.36$), reflecting the thermodynamic-electronic trade-off where stability and dense packing favor narrow gaps. FEPA correlates moderately with both Efinal ($\rho = 0.42$) and Ehull ($\rho = 0.48$), given their shared computational origin and stability implications. Notably, Ehull correlates with volume per atom ($\rho = 0.45$), indicating that metastable materials often feature looser atomic packing. Total magnetization remains largely independent ($|\rho| < 0.2$).

Takeaway 3. *Properties governed by related physical mechanisms exhibit stronger label-space correlations, while physically distinct properties remain weakly coupled. Moreover, the correlation patterns vary across datasets, necessitating consideration of dataset-specific statistical structure.*

4.2 Grouping Motivation and Algorithm

To better leverage inter-property relationships, we propose a hierarchical grouping strategy that identifies task structure before model

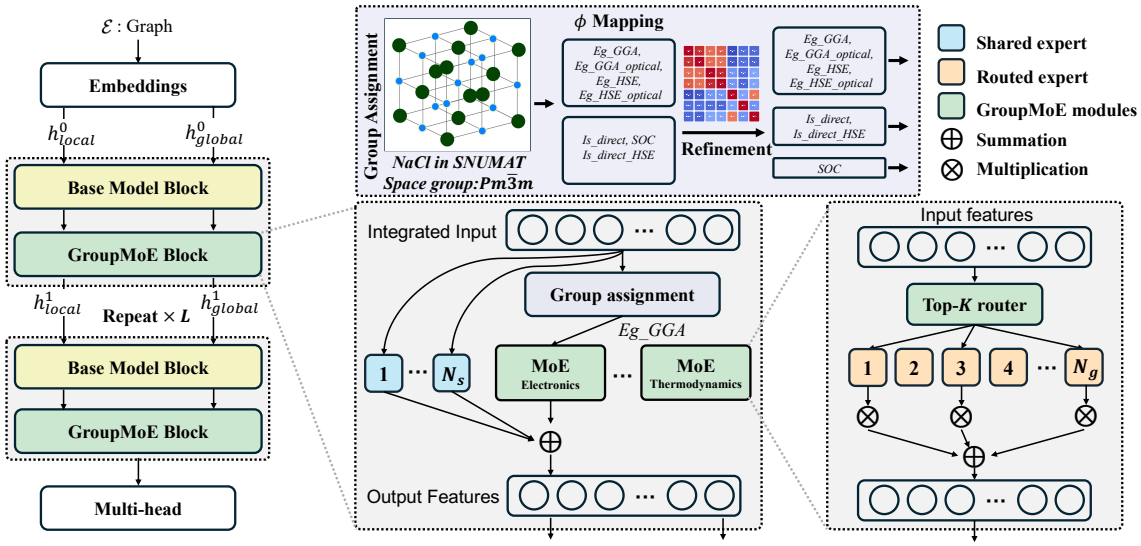


Figure 2: GroupMoE Architecture. (Left) Model architecture with alternating base model blocks, such as PotNet, ComFormer, or ReciNet, and GroupMoE blocks. (Top) 7-task grouping example for NaCl in the SNUMAT dataset. (Center) Group-specific MoE modules with shared experts. (Right) Top-K routing strategy combining expert outputs via weighted aggregation. Overall, properties are routed to group-specific MoEs based on group assignment, enabling specialized knowledge while maintaining shared representations.

training. Most prior MTL and MoE studies focus on domains such as vision or language, where task relationships are often ambiguous or purely empirical [6, 21, 31]. To estimate task relationships, some methods rely on gradient-based measures, but computing K gradients per step introduces substantial overhead [28]. Others require training a full model before measuring task similarity, which is impractical for large-scale applications [14]. In contrast, Section 4.1 shows that crystalline property prediction is inherently more structured: task grouping information can be determined from domain knowledge and label correlation analysis alone, before training.

Therefore, we propose a *hierarchical grouping algorithm* (Algorithm 1) that automatically constructs the task partition by coupling a domain-knowledge prior with data-driven statistical refinement. Formally, the algorithm realizes a mapping $\mathcal{A} : (\phi, \{y_t\}_{t\in\mathcal{T}}, \bar{\tau}, \underline{\tau}) \mapsto (\Pi, \mathcal{U})$ that, for any task set $\mathcal{T} = \{1, \ldots, T\}$, returns a partition $\Pi = \{\mathcal{G}_1, \ldots, \mathcal{G}_K\}$ together with an ungrouped set $\mathcal{U} \subseteq \mathcal{T}$, where K is the number of groups and each member of $\mathcal{U}$ forms its own singleton group. Its two stages operate at different levels of the knowledge hierarchy: Stage I instantiates a coarse partition from the standard physical-chemical taxonomy of material properties, and Stage II refines this partition against the empirical evidence carried by the training labels. The algorithm is *dataset-agnostic and training-free*: its only inputs are the category prior ϕ and the training labels. Applying it to a new dataset is therefore fully mechanical, and the group assignment is determined by Algorithm 1 rather than chosen by hand.

Stage I: Physical-Chemical Initialization. The algorithm begins by categorizing properties based on their underlying physical-chemical principles and types. Let $\phi : \mathcal{T} \to C$ map each task to its property category, inducing the initial partition $\Pi_0 = \{\mathcal{G}_c \mid c \in C\}$ with $\mathcal{G}_c = \{t \in \mathcal{T} \mid \phi(t) = c\}$. For crystalline property prediction, ϕ is initialized as (i) *electronic*, including band gaps computed via different DFT functionals (Eg_MBJ, Eg_OPT, Eg_GGA, Eg_HSE); (ii) *thermodynamic*, such as formation energy (FEPA), total energy (Etotal), and energy above hull (Ehull); (iii) *structural*, including density and VPA; and (iv) *magnetic*, such as total magnetization (M). Tasks within the same category probe related aspects of the electronic or atomic structure and are expected to benefit from

Algorithm 1 Hierarchical Grouping

Require: Tasks $\mathcal{T} = \{1, \ldots, T\}$, training labels $\{y_t\}_{t\in\mathcal{T}}$, category map $\phi : \mathcal{T} \to C$, thresholds $\bar{\tau}$ and $\underline{\tau}$
Ensure: Partition Π and ungrouped set $\mathcal{U}$

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Initialize partition by physical-chemical prior: Π_0 ← { G_c = {t ∈ T | φ(t) = c} | c ∈ C }
Compute pairwise masked Pearson correlations ρ_tj from training labels via eq. (2)
Construct statistical affinity matrix A_tj ← |ρ_tj|
Initialize Π ← Π_0, U ← ∅
for each task t ∈ T do
    Let G_c denote the current group of t in Π
    Compute in-group support a_in(t) = max_{j∈G_c\{t}} A_tj
    Compute out-of-group support a_out(t) = max_{j∉G_c} A_tj
    if a_in(t) ≥ τ̄ then
        Keep t in G_c
    else if a_out(t) ≥ τ then
        Reassign t to the group containing arg max_{j∉G_c} A_tj
    else
        Mark t as ungrouped: U ← U ∪ {t} and remove t from Π
    end if
end for
return Partition Π and ungrouped set U
```

shared representations. Crucially, ϕ depends only on universally defined property semantics in material science, and is therefore reusable across datasets.

Stage II: Label-Space Correlation Refinement. Knowledge priors alone can miss dataset-specific statistical interactions, so the second stage refines Π_0 against the training labels. The algorithm constructs the statistical affinity matrix $A \in [0,1]^{T\times T}$ with $A_{tj} = |\rho_{tj}|$, where ρ_{tj} follows eq. (2) and is estimated under pairwise non-missing label masks. For each task $t \in \mathcal{G}_c \in \Pi_0$, it measures the strongest in-group and out-of-group affinities,

$$a_{\text{in}}(t) = \max_{j\in\mathcal{G}_c\setminus\{t\}} A_{tj}, \qquad a_{\text{out}}(t) = \max_{j\notin\mathcal{G}_c} A_{tj}, \tag{3}$$

and applies a dual-threshold reassignment rule: t stays in $\mathcal{G}_c$ if $a_{\text{in}}(t) \geq \bar{\tau}$; otherwise, when $a_{\text{out}}(t) \geq \underline{\tau}$. t migrates to the group containing its strongest out-of-group partner $\arg\max_{j\notin\mathcal{G}_c} A_{tj}$ and t joins $\mathcal{U}$ when neither condition holds. Reassignment thus requires both weak in-group support and a clearly stronger out-of-group alternative. Rather than tuning $\bar{\tau}$ and $\underline{\tau}$ as opaque hyperparameters, we treat them as interpretable cutoffs on correlation magnitude and reuse $\bar{\tau} = 0.5$ and $\underline{\tau} = 0.4$ across all three benchmarks despite their different levels of correlation.

Beyond being training-free, Algorithm 1 is efficient and fully determined by its inputs with an $O(T^2S)$ correlation estimation over S training samples. The refinement is a single pass over tasks, so the partition for even the largest benchmark is produced in seconds, before any model training begins. The resulting assignments are summarized in Table 6. We also validate the algorithm empirically : the ablation in Section 5.2 shows that each stage contributes to final accuracy, and the sensitivity analysis in Table 10 shows that performance is robust over a reasonable threshold interval.

4.3 Model Architecture

To effectively leverage the grouping information obtained by Algorithm 1, we further propose **GroupMoE**, which explicitly incorporates the task grouping into the MoE architecture by constraining each property's routing to a group-specific sub-MoE (e.g., electronic or thermodynamic). The overall structure is illustrated in Figure 2 and consists of the following components.

Base Model with Plug-and-Play GroupMoE Blocks. Existing approaches for material property prediction predominantly adopt message-passing architectures, including both GNN-based and transformer-based models, to capture short- and long-range interactions across atomic structures, such as PotNet or ComFormer [18, 29, 39, 57]. However, these models typically do not support multi-property prediction or incorporate MoE mechanisms. To address this limitation, we introduce a modular GroupMoE block that can be seamlessly integrated into existing message-passing pipelines in a plug-and-play manner. Suppose the output of the base model block at the l-th layer for task t with input feature h_t^l is denoted as $z_t^l = \text{BaseModelBlock}(h_t^l)$, where $h_t^l = (h_{\text{local},t}^l, h_{\text{global},t}^l)$ represents the concatenated local and global input features. The GroupMoE block is inserted after each message-passing layer and operates on the intermediate representations z_t^l. After L rounds of message passing, a multi-head decoder outputs task-specific property predictions used for computing the final loss.

GroupMoE Block. Given the intermediate structural representations z_t^l from the base model block, we process each task individually through the GroupMoE block. Specifically, each task is first assigned to the group-specific MoE module (e.g., electronics or thermodynamics) according to the group assignment determined in Section 4.2. The task's representations z_t^l are then passed to the chosen MoE module and to the shared experts that are shared across all MoE modules. Each MoE module inherits the standard sparse MoE structure [48], and the final output h_t^{l+1} of the l-th message-passing layer for task t is

$$h_t^{l+1} = \underbrace{\sum_{i=1}^{N_g} \mathcal{R}_i^g(z_t^l) E_i^g(z_t^l)}_{\text{group-specific experts}} + \underbrace{\frac{1}{N_s}\sum_{j=1}^{N_s} E_j^s(z_t^l)}_{\text{shared experts}}, \tag{4}$$

where N_g denotes the number of group-specific experts, $\mathcal{R}^g$ denotes the same routing network with the $\text{Top}K(\cdot)$ operator as in eq. (1) for a specific group g, E_i^g denotes a group-specific expert, N_s denotes the number of shared experts, and E_j^s represents the j-th shared expert network. Importantly, the shared experts $E_j^s, \forall j \in [N_s]$, are always activated and are designed to capture generalizable features that benefit all task groups, promoting positive transfer and serving as a consistent representational backbone [30]. In contrast, the group-specific experts focus on specialized signals tailored to the properties within each group. Currently, the outputs of shared and group-specific experts are combined with equal weight, as the primary focus is on the effect of task grouping. As a potential extension, one could introduce a learnable weighting scheme to dynamically adjust their relative contributions per property. Lastly, this architecture recovers the sparse MoE by removing shared experts ($N_s = 0$) and allocating all tasks to a single group: GroupMoE is thus a strict generalization of the sparse MoE baseline studied in Section 3, and the difference between the two isolates precisely the effect of the injected domain knowledge.

5 Experimental Analysis of GroupMoE

In the context of **RQ3**, we comprehensively evaluate GroupMoE in terms of predictive performance, backbone generalization, training efficiency, and expert specialization. We further conduct design ablations to examine the contributions of hierarchical grouping, shared experts, and grouping thresholds.

5.1 Does Knowledge-Informed Routing Work?

Main results. Tables 2 and 3 (and Table 9 in the appendix) show that GroupMoE achieves the lowest $\Delta m\%$ on all three datasets: 1.92 on SNUMAT, 25.60 on JARVIS-DFT, and 35.99 on Materials Project. The most controlled comparison is against sparse MoE, which shares the architecture family and differs only in the absence of grouped routing and shared experts: GroupMoE improves $\Delta m\%$ from 4.94 to 1.92 on SNUMAT, from 39.45 to 25.60 on JARVIS-DFT, and from 48.51 to 35.99 on Materials Project, turning the weakest method in the study (on Materials Project) into the strongest. GroupMoE also clearly outperforms the best generic baseline on each dataset: UW on SNUMAT (3.51) and Materials Project (36.87), and MGDA on JARVIS-DFT (35.66).

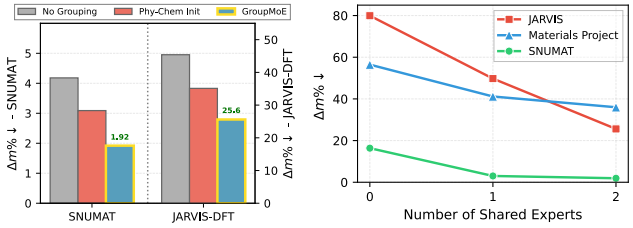


Figure 3: Ablation study on grouping strategies (left) and the shared experts (right). Lower $\Delta m\%$ is better.

Table 4: GroupMoE as a plug-and-play module measuring $\Delta m\%$ across base models.

Dataset	SchNet [45]	CGCNN [56]	Matformer [58]	PotNet [29]	eComFormer [57]	ReciNet [39]
SNUMAT	15.22	11.83	8.49	7.62	6.21	**1.92**
JARVIS-DFT	77.28	58.30	51.28	47.73	35.05	**25.60**

Generalization across Backbones. We examine backbone dependence from two complementary views. *First, varying the backbone under a fixed method* (Table 4): performance varies based on the underlying model capacity. SchNet and CGCNN only capture short-range interactions, showing relatively high $\Delta m\%$ values (e.g., 11.83 and 15.22 on SNUMAT, respectively). In contrast, PotNet and eComFormer benefit from long-range interaction modeling and achieve lower $\Delta m\%$ (e.g., 51.28 and 47.73 on JARVIS-DFT). ReciNet achieves the best performance, due to the use of reciprocal-space information. *Second, varying the method under a fixed backbone* (Table 8): with eComFormer as the backbone on JARVIS-DFT, GroupMoE again attains the lowest $\Delta m\%$ (35.05), outperforming UW (42.71), MGDA (42.85), CAGrad (45.17), and LS (51.82).

Training Efficiency. GroupMoE achieves a favorable efficiency-performance trade-off (Figure 8). It is over 3× faster than training separate models per property and 1.5× faster than gradient manipulation methods, while incurring only 1.6–2.2× overhead compared to single-task learning.

Expert Specialization. We visualize the learned embeddings using t-SNE across all tasks for three datasets, which form distinct clusters aligned with hierarchical grouping (Figure 9). This visualization confirms that GroupMoE produces clearly separated task embeddings in representation space.

Takeaway 4. *GroupMoE pushes the efficiency-performance frontier by achieving the best overall accuracy while remaining substantially more efficient than separate per-property training as shown in Figure 1.*

Takeaway 5. *The proposed routing design is plug-and-play across both short-range and long-range material representation backbones, including GNN- and transformer-based architectures.*

5.2 Ablation Study: Which Design Choices Matter?

Importance of Grouping. As shown in Figure 3 (left), without grouping, the model suffers the largest performance drop, yielding the highest $\Delta m\%$ (4.18 on SNUMAT, 45.4 on JARVIS-DFT). Physical-chemical initialization alone improves performance by separating tasks based on domain knowledge, and the full hierarchical grouping, which further refines groups via label-space correlation, achieves the lowest $\Delta m\%$ of 1.92 and 25.60, respectively. Both stages are therefore essential.

Importance of Shared Experts. While group-specific experts capture specialized representations, shared experts enable knowledge transfer across groups. As shown in Figure 3 (right), removing shared experts leads to significant degradation across all datasets, with $\Delta m\%$ increasing by up to 51% on JARVIS-DFT. This suggests that, despite being assigned to different groups, tasks still benefit from common representations, likely capturing universal structural or chemical patterns.

Grouping Thresholds. We provide additional analysis of the grouping thresholds τ in Table 10 in the appendix. Specifically, these thresholds act as interpretable cutoffs derived from the statistical task-correlation structure rather than arbitrary hyperparameters, and results on JARVIS demonstrate that the method remains robust across different threshold choices.

Takeaway 6. *Both grouping stages (physical initialization and correlation refinement) and the shared experts are necessary. The thresholds are interpretable correlation cutoffs.*

6 Limitations and Ethical Considerations

Limitations. Our evaluation focuses on crystalline materials, where inter-property relationships can often be supported by established physical principles. Extending GroupMoE to other scientific domains, such as molecular, catalytic, or biological protein property prediction, may require different forms of domain knowledge and grouping criteria tailored to their task structures.

Ethical Considerations. This work uses only publicly available materials science databases (SNUMAT, JARVIS-DFT, and MP) with no personal or human-subject data. The proposed methods reduce the computational cost of materials characterization, with potential positive societal impact through accelerated discovery of materials. We do not foresee direct negative societal impacts.

7 Conclusion

We present the first systematic study of multi-task learning for crystal multi-property prediction across three benchmarks, 18 properties, and diverse backbones. Our analysis reveals a clear efficiency-performance trade-off and shows that no generic MTL method performs consistently across datasets. While MoE provides favorable efficiency and scalability, its task-agnostic routing limits both performance and interpretability. To address this limitation, we analyze the physical-chemical principles plus label-space correlations, and then group the properties hierarchically. We incorporate this structure into the MoE routing to develop GroupMoE. It achieves the best overall performance on all three datasets while reducing training cost by more than 3× compared with separate per-property models. These results demonstrate the effectiveness of relationship-aware routing for scalable and interpretable multi-property prediction.

Generative AI Usage Statement

Generative AI tools were used to assist with language editing and manuscript restructuring. All research ideas, model designs, implementations, experiments, and analyses were conceived and conducted by the authors, taking full responsibility for the content.

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Table 5: Statistics of all datasets with physical meaning description and task type. It is important to note that we retain the naming convention of each property from the original data source with the intent to provide the distribution of properties in each dataset.

Property	Task type	Physical Meaning Description	Dataset		
			SNUMAT	JARVIS-DFT	Materials Project
Bandgap	Regression	Electronic bandgap value	-	-	139,367
Bandgap (OPT)	Regression	Bandgap using OPT functional	-	55,722	-
Bandgap (MBJ)	Regression	Bandgap using MBJ functional	-	18,171	-
Bandgap GGA	Regression	Calculated fundamental band gap in GGA	10,481	-	-
Bandgap HSE	Regression	Calculated fundamental band gap in HSE	10,481	-	-
Bandgap GGA Optical	Regression	Calculated direct band gap in GGA	10,481	-	-
Bandgap HSE Optical	Regression	Calculated direct band gap in HSE	10,481	-	-
FEPA	Regression	Formation energy per atom	-	55,722	139,367
Efinal	Regression	Final energy per atom	-	-	139,367
Etotal	Regression	Total energy per atom	-	55,722	-
Ehull	Regression	Energy above hull	-	55,370	139,285
Efermi	Regression	Fermi level energy	-	-	139,309
Density	Regression	Mass density	-	-	139,367
VPA	Regression	Volume per atom	-	-	139,367
M	Regression	Total magnetization per formula	-	-	139,367
Is Direct	Classification	Type of band gap in GGA (direct or indirect)	10,481	-	-
Is Direct HSE	Classification	Type of band gap in HSE (direct or indirect)	10,481	-	-
SOC	Classification	Spin-orbit coupling (True or False)	10,481	-	-

A Related Work

Material Property Prediction. Material property prediction has been extensively studied using both physics-based and deep-learning approaches. Traditional physics-based methods, such as Coulomb matrices [11, 44], effectively model ionic and metallic materials but struggle with generalization due to limitations such as permutation invariance. In contrast, deep learning enables more flexible modeling [16, 22, 52]. Subsequent efforts have shifted toward leveraging 3D geometric information, representing crystals as atomic graphs. SchNet [45] introduces continuous-filter convolution, while CGCNN [56] models periodicity via multi-edge graphs. ALIGNN [7] incorporates angle-based features, and Matformer [58] captures periodic patterns using self-connecting edges. PotNet [29] proposes physics-informed edge features with interatomic potentials. iComFormer and eComFormer [57] enhance representation through invariant descriptors and equivariant features. Crystalformer [50] models infinitely connected attention to encode periodicity, and CrystalFramer [18] learns SE(3)-invariant representations using dynamic atom-wise coordinate frames. ReciNet [39] introduces a crystal-native architecture that captures long-range interactions from learnable reciprocal space. Despite these advances, all of the above are designed for single-task learning: as our study in Section 3 confirms empirically, turning them into effective multi-property predictors is a non-trivial open problem rather than a straightforward application of existing MTL methods.

Multi-task Learning (MTL). MTL trains a single model to solve multiple tasks simultaneously, reducing computational cost. The standard approach, linear scalarization (LS), optimizes a weighted or averaged sum of task losses [5]. Although efficient, LS often suffers from imbalanced training as tasks converge at different rates. To alleviate this issue, Kendall et al. [24] weights individual loss functions based on homoscedastic uncertainties, while Liu et al. [32] adjusts weights based on the relative rate of loss descent to better synchronize task optimization. Another line of work formulates MTL as multi-objective optimization and employs the Multi-Gradient Descent Algorithm (MGDA) to mitigate gradient conflict [10, 46]. For example, CAGrad [31] constrains the update direction, while PCGrad [59] resolves conflicts by projecting gradients onto the normal plane. Nash-MTL [38] considers the gradient combination step as a bargaining game. SDMGrad [54] precisely controls the gradient conflict in each epoch with a direction-oriented mechanism, and FairGrad [3] incorporates α-fairness. Despite their effectiveness, gradient-based approaches incur high computational overhead because they require computing all task gradients. Our study (Section 3.2) additionally finds them to be the least stable family across materials datasets.

Mixture of Experts (MoE). The MoE framework was originally introduced to combine multiple local experts through conditional computation [19], but early variants are computationally expensive because all experts need to be activated. Sparse MoE [48] addresses this by selecting only a small subset of experts during training and inference, enabling scalability in NLP. MoE techniques have since been transferred to large models in computer vision and multimodal applications [37, 43, 53] with better routing strategies, and Mixture of Attention (MoA) integrates MoE into attention layers at low computational cost [60]. More recently, MoE has been adopted for MTL model design: M^3ViT [12] sparsely activates task-specific experts during training to avoid conflicts between tasks, and ModSquad [6] lets experts be both cooperative and specialized through a mutual-information loss. However, MoE-based MTL methods have seldom been explored in crystal property prediction, and their selected experts typically lack explicit roles for different tasks, limiting interpretability. Our study quantifies this gap (Section 3.2),

and GroupMoE addresses it by aligning expert allocation with scientifically grounded task groups (Section 4).

B Dataset and Analysis Details

B.1 Data Sources

SNUMAT. SNUMAT [25] contains 10,481 semiconducting inorganic materials with high-fidelity band gap information computed using both GGA and HSE functionals. It incorporates magnetic ordering and spin-orbit coupling (SOC) effects to improve accuracy over standard DFT approaches. Seven tasks are included: the fundamental and direct band gaps from GGA (Bandgap GGA, GGA Optical) and HSE (Bandgap HSE, HSE Optical) as regression tasks, and the band gap type classification (Is Direct, Is Direct HSE) and a binary SOC indicator (SOC) as classification tasks. Bandgap GGA and Bandgap HSE represent the fundamental band gaps in GGA and HSE (eV), while GGA Optical and HSE Optical correspond to the direct band gaps. Is Direct and Is Direct HSE classify the band gap type as direct or indirect under GGA and HSE, respectively, and SOC indicates the presence of spin-orbit coupling. These tasks span regression and classification objectives, reflecting key physical properties relevant to electronic and optoelectronic applications.

JARVIS-DFT. The JARVIS-DFT dataset, proposed by Choudhary et al. [8], contains 55,723 materials and serves as a benchmark for crystal property prediction. Evaluation includes five regression tasks: formation energy, total energy, bandgap (using both OPT and MBJ methods), and energy above the convex hull (Ehull). The training, validation, and testing splits for formation energy, total energy, and OPT bandgap consist of 44,578, 5,572, and 5,572 crystals, respectively. The splits for Ehull include 44,296, 5,537, and 5,537 crystals, while the MBJ bandgap tasks consist of 14,537, 1,817, and 1,817 crystals. The MBJ functional is considered more accurate for bandgap calculations compared to OPT, and both are utilized in this study. Notably, 18,865 of the dataset's crystal structures have been experimentally observed, adding robustness to its use in predictive modeling.

Materials Project. The Materials Project (MP) database [20] is an online public materials database providing various materials structures and their properties. It supports API access for automated data retrieval and has been instrumental in accelerating the discovery of computational materials. We specifically use its snapshot collected by Gong et al. [15], which contains 139,367 materials.

B.2 Dataset Distributions

We present the statistical distributions of material properties across the three benchmark datasets used in our experiments. All distributions are computed from the combined train, validation, and test splits to provide a comprehensive view of the dataset characteristics.

SNUMAT. Figure 5 presents distributions for materials with four bandgap calculations using different functionals (GGA and HSE) and approximations (fundamental vs. optical), along with three electronic structure classifications. The fundamental bandgaps (μ_{GGA} = 1.77 eV, μ_{HSE} = 2.86 eV) are systematically lower than optical bandgaps (μ_{GGA} = 2.40 eV, μ_{HSE} = 3.53 eV), reflecting the excitonic effects captured in optical calculations. Classification distributions reveal that 71.8% of materials exhibit indirect bandgaps under HSE calculations, while only 1.2% show spin-orbit coupling effects.

JARVIS-DFT. Figure 6 displays distributions for materials from the Joint Automated Repository for Various Integrated Simulations. Formation energies are centered at −0.83 eV/atom with moderate spread (σ = 1.08 eV/atom). The OptB88vdW bandgap distribution (μ = 0.69 eV) shows a pronounced peak near zero, indicating a prevalence of metallic or small-gap materials. Energy above hull values (μ = 1.78 eV/atom) suggests the inclusion of more metastable structures. The MBJ bandgap subset represents materials with available meta-calculations.

Materials Project. Figure 7 shows the distribution of eight properties from the Materials Project dataset. The dataset exhibits diverse property ranges: band gaps are predominantly concentrated below 2 eV (μ = 1.06 eV), formation energies show a symmetric distribution centered at −1.42 eV/atom, and magnetic moments display a heavily right-skewed distribution with most materials being non-magnetic. Energy above hull values, indicating thermodynamic stability, is concentrated near zero with an exponential decay. All distributions display characteristic patterns of databases: formation energy distributions approximate Gaussian shapes centered at negative values, bandgap distributions are right-skewed with peaks near zero (reflecting metallic materials), and stability metrics (energy above hull) follow exponential decay patterns favoring thermodynamically stable configurations.

B.3 Inter-Task Relationship Analysis Details

Unlike conventional multi-task learning scenarios such as image classification, where tasks typically share similar semantic structures, tasks in crystal property prediction often differ significantly in their underlying chemical mechanisms and structure-property mappings. Each task corresponds to a distinct property (e.g., band gap, formation energy, total energy), leading to heterogeneous functional relationships between crystal structures and tasks. To systematically quantify the potential for information sharing, we analyze the label-space correlation via pairwise Pearson correlation coefficients on the training labels across the three datasets, exposing *a priori* statistical dependencies that identify tasks statistically primed for effective joint learning (Figure 4). Here we provide the detailed analysis complementing Section 4.1.

B.3.1 Physical-Chemical Relationships.

SNUMAT. This dataset focuses exclusively on electronic bandgap properties computed under varying density functional approximations and physical definitions, creating hierarchical task relationships. Electronic bandgaps computed with GGA and HSE functionals measure identical physical quantities, the minimum energy difference between valence and conduction band edges, but treat electron-electron interactions through different density functional approximations. The distinction between electronic and optical bandgaps arises from momentum conservation in optical transitions: the electronic gap is the minimum energy difference between valence and conduction bands which may occur at different k-points (indirect gap), while the optical gap is the minimum energy for a direct, vertical optical transition at a single k-point, as photons carry negligible momentum [40]. The classification of bandgap

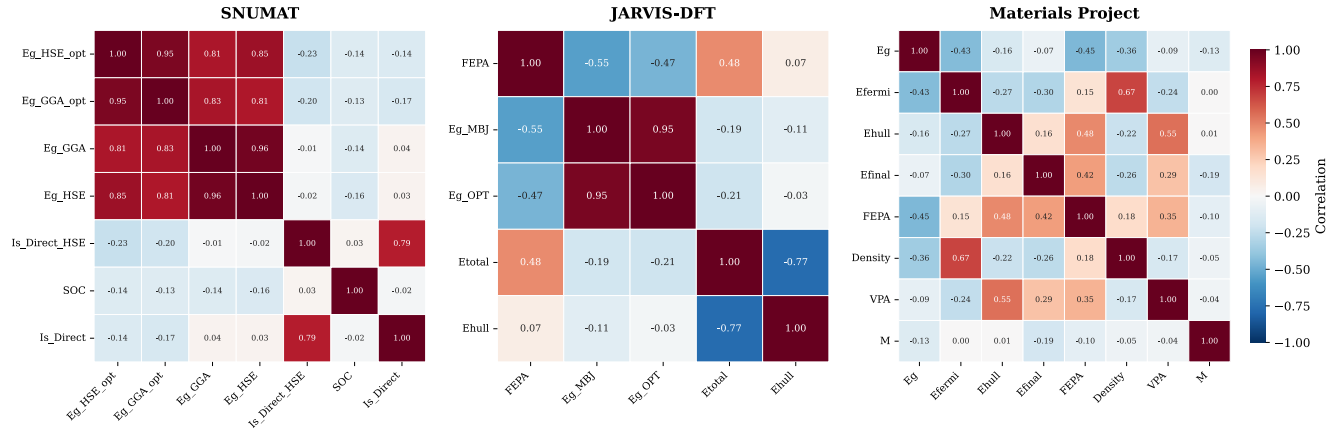


Figure 4: Pearson correlation matrices revealing label-space correlation across three datasets, SNUMAT (7 tasks), JARVIS-DFT (5 tasks), and Materials Project (8 tasks). Strong correlations indicate tasks suitable for shared representations in multi-task learning, which guides task grouping in our framework. For instance, in SNUMAT, four bandgap properties exhibit strong positive correlation ($\rho \geq 0.81$), as they represent the electronic band gap computed via different DFT functionals, probing the same underlying electronic structure.

type (direct versus indirect) is determined by crystal symmetry and atomic orbital character dictating band structure topology [47], with GGA and HSE largely agreeing on band topology due to their shared treatment of orbital occupations, though HSE's inclusion of exact Hartree-Fock exchange can occasionally shift band extrema positions. Spin-orbit coupling (SOC) represents a relativistic correction to the electronic Hamiltonian [27] whose magnitude depends primarily on atomic number rather than bandgap characteristics, explaining its statistical independence from other electronic properties.

JARVIS-DFT. The five properties exhibit differences rooted in electronic structure and thermodynamic principles. Formation energy and bandgap represent fundamentally anti-correlated quantities: materials with favorable formation energies typically exhibit metallic or narrow-gap character due to electronic delocalization, while large-gap insulators sacrifice stability for localization [9, 41]. The MBJ and OptB88vdW bandgaps, despite different theoretical foundations (Becke-Johnson potential versus van der Waals corrections), probe the same Kohn-Sham energy gap [26, 51], yielding consistent electronic structure features. Total energy and Ehull are coupled through thermodynamic stability: materials with lower total energies naturally exhibit smaller decomposition tendencies (lower hull energies), as energetically favorable structures reside near the convex hull [49]. However, Ehull (considering all competing phases) encodes information largely independent of a material's intrinsic thermodynamic favorability (formation energy) and electronic structure (bandgaps), explaining their statistical independence.

Materials Project. This dataset encompasses eight properties spanning electronic, energetic, structural, and magnetic domains, creating a rich multi-physics task structure. Bandgap and formation energy per atom exhibit the inverse relationship also observed in JARVIS: thermodynamically favorable materials tend toward metallic or narrow-gap behavior, while wide-gap insulators sacrifice stability for electronic localization [41]. The Fermi energy (Efermi), representing the electrochemical potential of electrons, couples to atomic packing: materials with high electron densities exhibit elevated Fermi levels due to Pauli exclusion filling higher energy states [1]. The energetic properties, formation energy per atom (FEPA), final energy per atom (Efinal), and energy above hull (Ehull), form a conceptually related triad measuring different thermodynamic aspects. Formation energy measures stability relative to constituent elements, final energy represents absolute DFT total energy, and hull energy quantifies metastability. Ehull shows how far a material's energy lies above the thermodynamic stability line (convex hull), where Ehull = 0 indicates a stable phase and Ehull > 0 indicates a thermodynamic tendency to decompose into lower-energy competing phases [20]. Finally, total magnetization represents a quantum phenomenon from unpaired electron spins in transition metal compounds, showing weak correlations with other properties due to its material-class specificity (predominantly transition metals with partially filled d-orbitals).

B.3.2 Label-Space Correlation. We compute pairwise Pearson correlation coefficients between training labels within each dataset via

$$\rho_{ij} = \frac{\sum_{s=1}^{S}(y_i^s - \bar{y}_i)(y_j^s - \bar{y}_j)}{\sqrt{\sum_{s=1}^{S}(y_i^s - \bar{y}_i)^2}\sqrt{\sum_{s=1}^{S}(y_j^s - \bar{y}_j)^2}}, \tag{5}$$

where y_i^s denotes the label of property i for sample s and $\bar{y}_i = \frac{1}{S}\sum_{s=1}^{S} y_i^s$.

SNUMAT. The label-space correlations reveal clear hierarchical clustering aligned with physical-chemical principles. Electronic and optical bandgaps computed with GGA and HSE show exceptional correlation separately ($\rho = 0.96$ for electronic and $\rho = 0.95$ for optical), indicating that while HSE systematically predicts larger

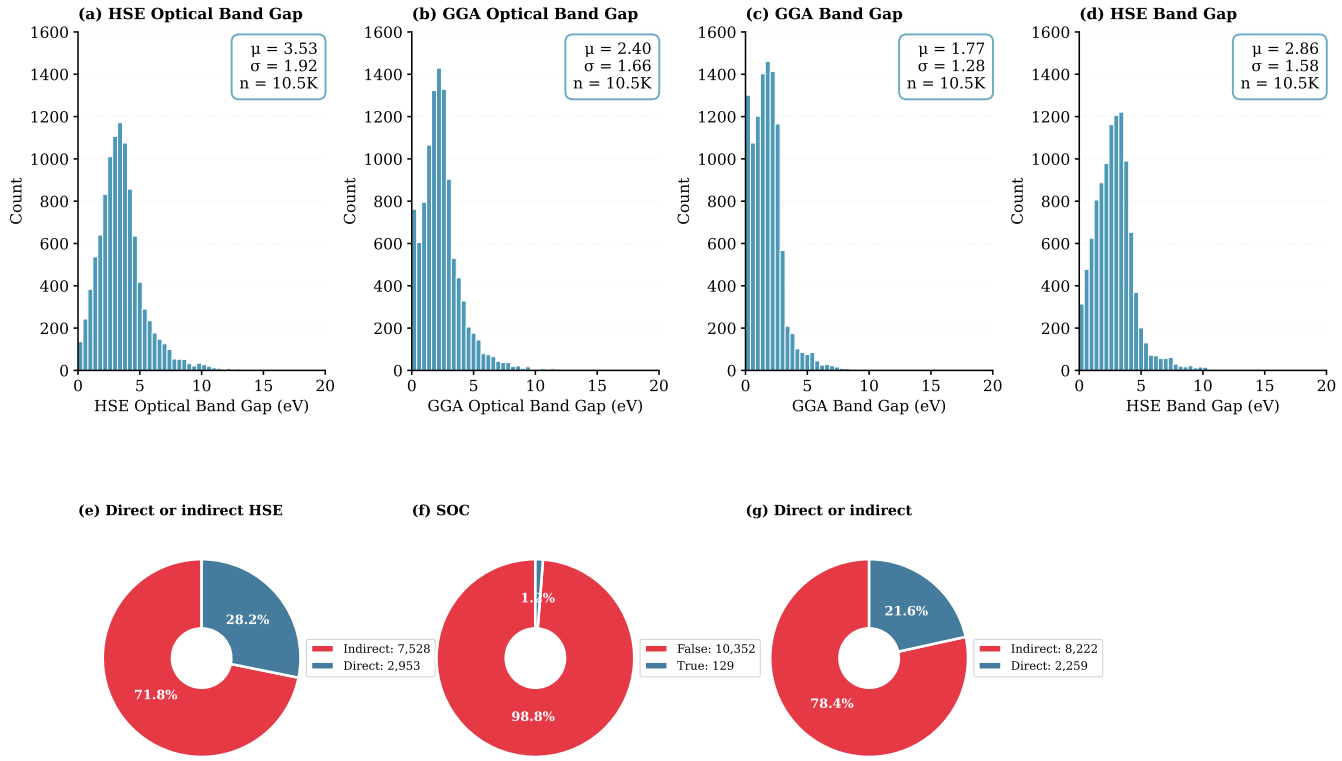


Figure 5: Distribution of material properties in the SNUMAT dataset. (a-d) Histograms show bandgap values computed using GGA and HSE functionals for both fundamental and optical transitions. (e-g) Pie charts display classification distributions for direct/indirect bandgap character under HSE functional, spin-orbit coupling (SOC) effects, and direct/indirect character under GGA functional. Statistics boxes report mean (μ), standard deviation (σ), and sample count (n).

gaps (typically 0.5-1 eV higher due to reduced self-interaction error [17]), both functionals yield nearly identical rankings of materials by gap magnitude. The four regression tasks (GGA/HSE electronic/optical gaps) form a strongly correlated cluster with pairwise correlations exceeding $\rho = 0.81$; within this cluster, optical and electronic gaps show strong correlations ($\rho = 0.81$-0.85), with systematic deviations arising from momentum conservation constraints: direct-gap materials exhibit nearly identical optical and electronic gaps, while indirect-gap materials show larger optical gaps due to vertical transition requirements. The direct/indirect classifications show strong cross-functional agreement ($\rho = 0.79$), confirming that GGA and HSE largely agree on band topology. In contrast, optical gap magnitude shows weak negative correlation with direct/indirect classification ($\rho = -0.20$ to -0.23), indicating that gap type is largely independent of gap magnitude, though direct-gap materials show a slight tendency toward smaller optical gaps. Notably, SOC shows near-zero correlations with all properties ($|\rho| < 0.16$), confirming its statistical independence from bandgap characteristics and motivating dedicated task-specific treatment.

JARVIS-DFT. The observed correlations align with physical-chemical expectations. The MBJ and OptB88vdW bandgaps exhibit near-perfect correlation ($\rho = 0.95$), confirming that despite different DFT corrections, both functionals capture consistent electronic structure features and yield similar material information. Total energy and hull energy exhibit a strong negative correlation ($\rho = -0.77$), confirming that energetically favorable structures (lower total energies) tend to reside near the convex hull (smaller hull energies), consistent with thermodynamic stability principles. In contrast, Ehull shows near-zero correlations with formation energy ($\rho = 0.07$) and both bandgaps ($\rho \approx -0.1$), as hull energy measures stability relative to competing phases in the local composition space, independent of thermodynamic favorability (formation energy) and electronic structure (bandgaps).

Materials Project. The correlation structure across eight properties reveals distinct task clusters aligned with physical-chemical relationships. The Fermi energy-density correlation ($\rho = 0.67$) represents the strongest relationship in the dataset, validating that electron chemical potential scales with atomic packing density

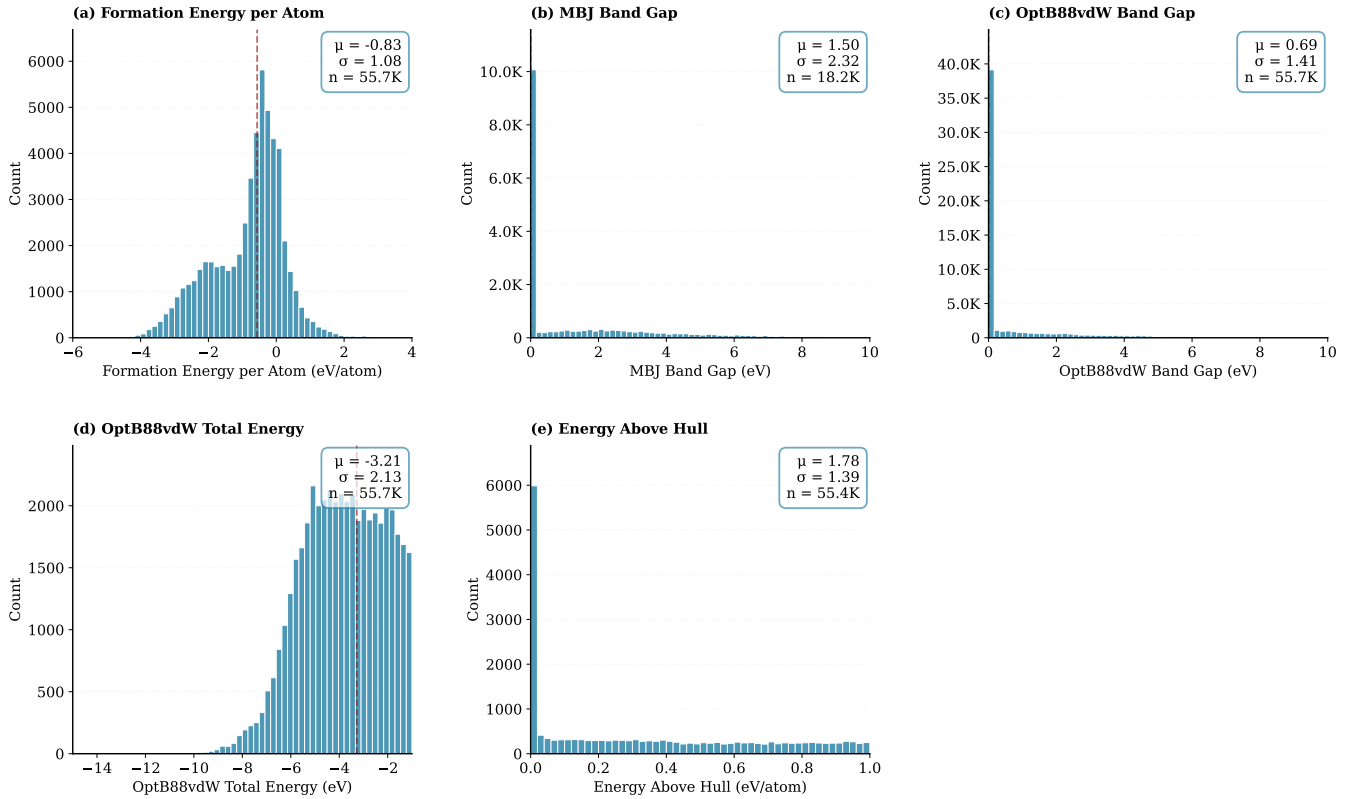


Figure 6: Distribution of material properties in the JARVIS-DFT dataset. (a) Formation energy per atom, (b) MBJ bandgap, (c) OptB88vdW bandgap, (d) OptB88vdW total energy per atom, and (e) energy above convex hull. Red dashed lines indicate median values.

through Pauli exclusion principles, motivating their joint representation. Bandgap exhibits moderate negative correlations with formation energy ($\rho = -0.45$), Fermi energy ($\rho = -0.43$), and density ($\rho = -0.36$), confirming the thermodynamic-electronic trade-off where stable, dense materials favor narrow-gap behavior while sacrificing stability. Within energetics, formation energy shows moderate correlations with final energy ($\rho = 0.42$) and hull energy ($\rho = 0.48$): final energy and formation energy share computational origins (formation energy derived from final energy calculations), while formation energy and hull energy both measure thermodynamic stability from different reference frames (elements versus competing phases). Hull energy shows moderate correlation with volume per atom ($\rho = 0.45$), revealing metastability sensitivity to atomic packing. Total magnetization shows near-zero correlations with all properties ($|\rho| < 0.2$), confirming its material-class specificity and motivating dedicated task-specific treatment.

C Experimental Details

C.1 Hyperparameter Settings

We provide detailed hyperparameter configurations for GroupMoE across the three benchmark datasets. For all datasets, we train for 500 epochs using the AdamW optimizer with a weight decay of 10^{-4}. The batch size is set to 64 for SNUMAT and JARVIS-DFT, and 128 for Materials Project due to its larger dataset size. The number of task groups is determined by our grouping strategy: 3 groups for SNUMAT, 2 groups for JARVIS-DFT, and 4 groups for Materials Project. For the mixture-of-experts architecture, we use 6 experts per group with top-$K = 3$ routing, meaning each input activates 3 experts from the assigned group. Additionally, we employ 2 shared experts across all groups to capture common materials knowledge. The learning rate varies by dataset: 6×10^{-4} for SNUMAT, 1×10^{-3} for JARVIS-DFT, and 2×10^{-4} for Materials Project, selected based on preliminary validation performance. These hyperparameters are summarized in Table 7. We note that these settings were chosen based on a limited hyperparameter search, primarily focusing on learning rate and the number of experts. Further improvements may be achievable through more extensive tuning, such as adjusting the

Table 6: Grouping assignments across datasets.

Group	SNUMAT	JARVIS-DFT	Materials Project
G1	Eg_GGA, Eg_HSE, Eg_GGA_optical, Eg_HSE_optical	Eg_MBJ, Eg_OPT, FEPA	Eg, Efinal, FEPA
G2	Is_direct, Is_direct_HSE	Etotal, Ehull	Efermi, Density
G3	SOC	–	Ehull, VPA
G4	–	–	M

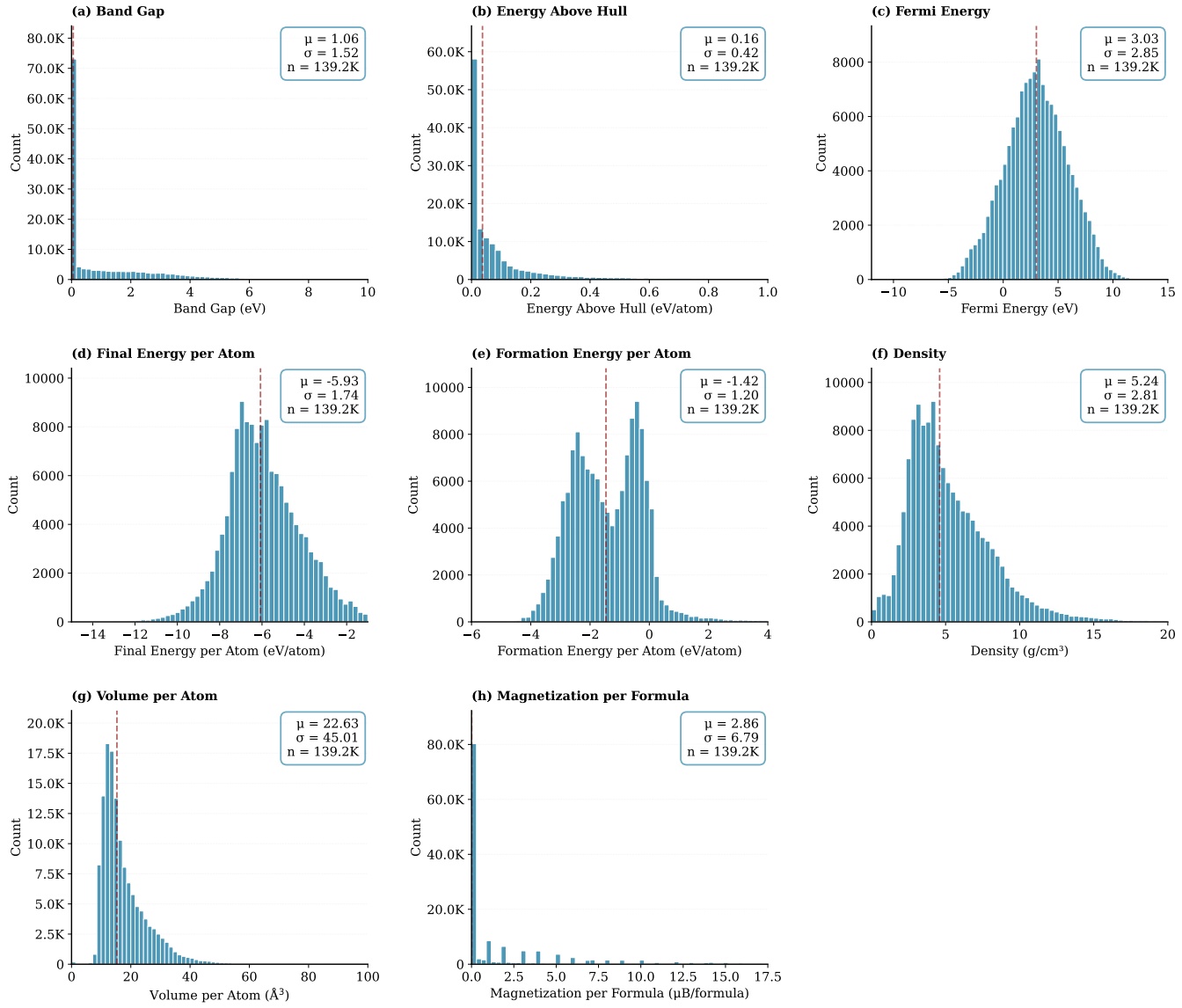


Figure 7: Distribution of material properties in the Materials Project dataset. (a) Band gap, (b) energy above convex hull, (c) Fermi energy, (d) final energy per atom, (e) formation energy per atom, (f) density, (g) volume per atom, and (h) total magnetization per formula unit. Red dashed lines indicate median values.

Table 7: Model settings of GroupMoE for SNUMAT, JARVIS-DFT, and the Materials Project datasets.

	# Groups	Total epochs	Learning rate	Shared experts	# Experts per group	Top-k
SNUMAT	3	500	6e-4	2	6	3
JARVIS-DFT	2	500	1e-3	2	6	3
Materials Project	4	500	2e-4	2	6	3

number of experts per group, the top-K value, or the auxiliary loss coefficient for load balancing.

Table 8: Performance of all MTL methods with eComFormer as the backbone on the JARVIS-DFT dataset.

Methods	FEPA↓	Bandgap (MBJ)↓	Bandgap (OPT)↓	Etotal↓	Ehull↓	$\Delta m\%$↓
	eV/atom	eV	eV	eV/atom	eV	
eComFormer+LS	0.0405	0.2184	0.1182	0.0676	0.0833	51.82
eComFormer+UW	0.0373	0.2230	0.1184	0.0616	0.0768	42.71
eComFormer+MGDA	0.0399	0.2107	0.1124	0.0604	0.0794	42.85
eComFormer+CAGrad	0.0403	0.2057	0.1114	0.0629	0.0812	45.17
eComFormer+GroupMoE (**ours**)	0.0366	0.2232	0.1156	0.0543	0.0746	**35.05**

Table 9: Performance comparison on the Materials Project dataset. Regression tasks are evaluated using the test MAE (lower is better).

Methods	Bandgap↓	Efermi↓	Ehull↓	Efinal↓	FEPA↓	Density↓	VPA↓	M↓	$\Delta m\%$↓
	eV	eV	eV/atom	eV/atom	eV/atom	g/cm^3	Å^3/atom	μ_B/f.u.	
STL	0.206	0.320	0.024	0.042	0.020	0.031	0.315	1.053	
LS	0.196	0.317	0.026	0.061	0.032	0.066	0.669	1.090	42.08
UW	0.191	**0.314**	0.025	0.062	**0.031**	**0.058**	0.619	1.094	36.87
MGDA	0.190	0.321	0.025	0.069	0.033	0.064	0.609	1.055	44.50
CAGrad	0.192	0.319	0.024	0.063	0.034	0.066	0.669	1.067	43.75
Sparse MoE	0.194	0.319	0.025	0.061	0.033	0.063	0.831	1.082	48.51
GroupMoE (**ours**)	**0.188**	0.324	**0.024**	**0.058**	0.036	0.065	**0.499**	**1.045**	**35.99**

C.2 Results on Materials Project

On the Materials Project dataset (Table 9), GroupMoE obtains the best overall performance with a $\Delta m\%$ of 35.99, surpassing UW (36.87) and significantly improving over sparse MoE (48.51). It achieves the best results on 5 out of 8 tasks, including bandgap, Ehull, Efinal, VPA, and M.

C.3 Grouping Threshold Sensitivity

The grouping procedure relies on two thresholds, $\bar{\tau}$ and $\underline{\tau}$, which control the in-group and out-of-group affinity cutoffs during the label-space refinement stage. Rather than tuning these as opaque hyperparameters, we set them directly from the statistical correlation structure observed in Section 4.1, treating them as interpretable cutoffs on Pearson correlation magnitude. Table 10 reports performance on JARVIS-DFT under three threshold settings spanning a representative range. Across all configurations, GroupMoE attains $\Delta m\%$ values between 25.60 and 35.11, all of which substantially outperform every baseline reported in the main results (Table 3). The best performance is obtained at $(\bar{\tau}, \underline{\tau}) = (0.5, 0.4)$, which we adopt throughout the main experiments. This robustness indicates that the grouping mechanism is governed by the underlying correlation structure rather than by precise threshold values, and the thresholds need not be finely tuned to recover competitive performance.

C.4 Training Time Comparison

Figure 8 compares the training time per epoch across different MTL methods, normalized relative to the single-task base model, ReciNet in this case. While GroupMoE incurs moderate overhead (1.72×, 1.57×, and 2.15× on SNUMAT, JARVIS-DFT, and Materials Project, respectively) compared to the base model, it is substantially more efficient than gradient manipulation methods such as MGDA and CAGrad, which require 2.46–4.49× longer training times due to per-task gradient computation. Importantly, GroupMoE is also over 3× faster than the naive approach of training multiple separate base models (one per task), while achieving superior performance. The computational overhead primarily stems from the mixture-of-experts routing mechanism.

Table 10: Performance comparison on five tasks from the JARVIS-DFT dataset with different thresholds.

$\bar{\tau}, \underline{\tau}$	FEPA↓	Bandgap (MBJ)↓	Bandgap (OPT)↓	Etotal↓	Ehull↓	$\Delta m\%$↓
	eV/atom	eV	eV	eV/atom	eV	
0.4, 0.4	0.048	0.213	0.125	0.041	0.075	35.11
0.5, 0.6	0.034	0.230	0.120	0.048	0.074	28.47
0.5, 0.4	0.034	0.213	0.112	0.047	0.072	25.60

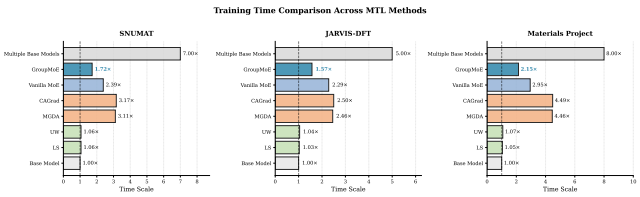


Figure 8: Training time per epoch relative to the single-task baseline (1.00×). GroupMoE achieves efficient multi-property prediction, outperforming both gradient-based methods and the naive approach of training separate models.

D Interpretability Analysis

To validate that GroupMoE learns meaningful task-specific representations aligned with physical property relationships, we perform interpretability analysis on the embedding features. Figure 9 presents t-SNE visualizations [34] of these learned embeddings across three diverse materials science datasets: SNUMAT (7 tasks, 3 groups), JARVIS-DFT (5 tasks, 2 groups), and Materials Project (8 tasks, 4 groups). For each dataset, we extract features after message-passing layers from the trained GroupMoE and apply t-SNE to visualize the high-dimensional task embeddings in 2D space. Tasks are colored individually based on their physical property categories, while gray lines denote the physics-based group boundaries defined by our grouping strategy.

The t-SNE projections reveal distinct clustering patterns that align with the prescribed task groups across all three datasets.

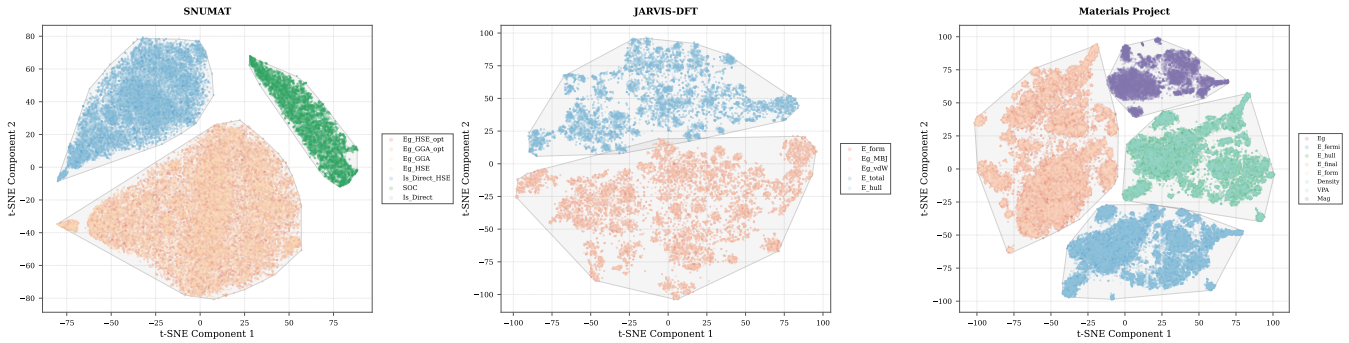


Figure 9: t-SNE visualization of learned task embeddings across three materials datasets. Each subplot shows the task embedding visualization result for different datasets, with points colored by task and gray lines indicating group boundaries. Left: SNUMAT (7 tasks, 3 groups) exhibits clear group separation, with bandgap regression tasks distinct from classification tasks and SOC. Middle: JARVIS-DFT (5 tasks, 2 groups) shows two primary clusters of formation/bandgap properties versus energy/stability metrics, with moderate overlap reflecting thermodynamic coupling. Right: Materials Project (8 tasks, 4 groups) demonstrates a clear four-group structure separating properties.

For SNUMAT, the four bandgap prediction tasks form a cohesive cluster in the embedding space, clearly separated from the direct/indirect classification tasks and the SOC task. Similarly, JARVIS-DFT exhibits two primary clusters: formation energy and bandgap tasks versus total energy and Ehull, though with a bit overlap. Materials Project demonstrates a clear four-group structure with bandgap/energy tasks, Fermi level/density properties, stability metrics, and magnetization with distinguishable regions. This visualization confirms that GroupMoE produces clearly separated task embeddings in representation space.